

Paper 2: Algorithms, programming and logic

Chapter 9: Databases

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Learning Objectives:

9 Databases

Candidates should be able to:

- 1 Define a single-table database from given data storage requirements
- 2 Suggest suitable basic data types
- 3 Understand the purpose of a primary key and identify a suitable primary key for a given database table
- 4 Read, understand and complete structured query language (SQL) scripts to query data stored in a single database table

Notes and guidance

- Including:
 - fields
 - records
 - validation
- Including:
 - text/alphanumeric
 - character
 - Boolean
 - integer
 - real
 - date/time
- Limited to:
 - SELECT
 - FROM
 - WHERE
 - ORDER BY
 - SUM
 - COUNT
- Identifying the output given by an SQL statement that will query the given contents of a database table

Resources:



<https://www.hoddereducation.co.uk/subjects/general/products/14-16/cambridge-igcse-and-o-level-computer-science-secon>

Resources:



Craig 'n' Dave For Students

CRAIG 'N' DAVE

To explore the entire series,
visit <https://craigndave.org/free-videos>

9.1 Define a single-table database from given data storage requirements

Click the link below to watch the video for Database overview

<https://www.youtube.com/watch?v=hy7FwqJyE6c>

9 Databases

Database introduction

CAMBRIDGE
IGCSE
(0478/0984)



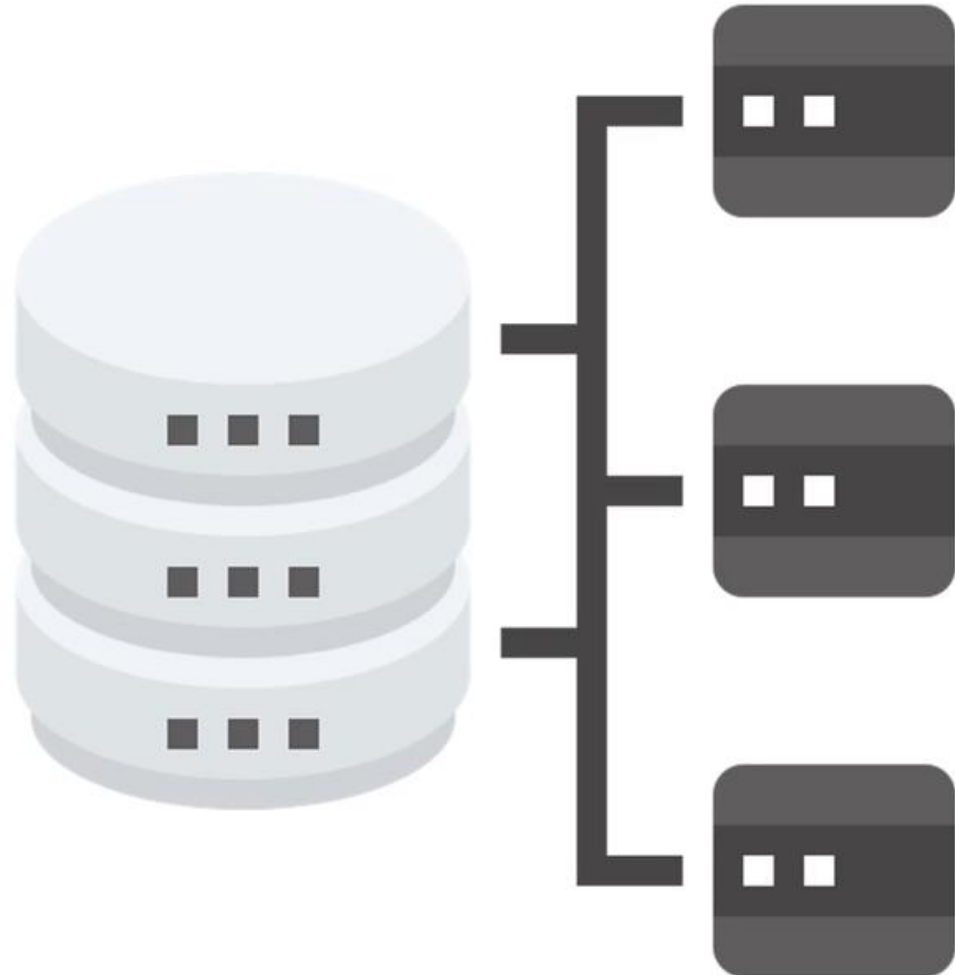
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9.1 Define a **single-table database** from given **data storage** requirements

What is a database?

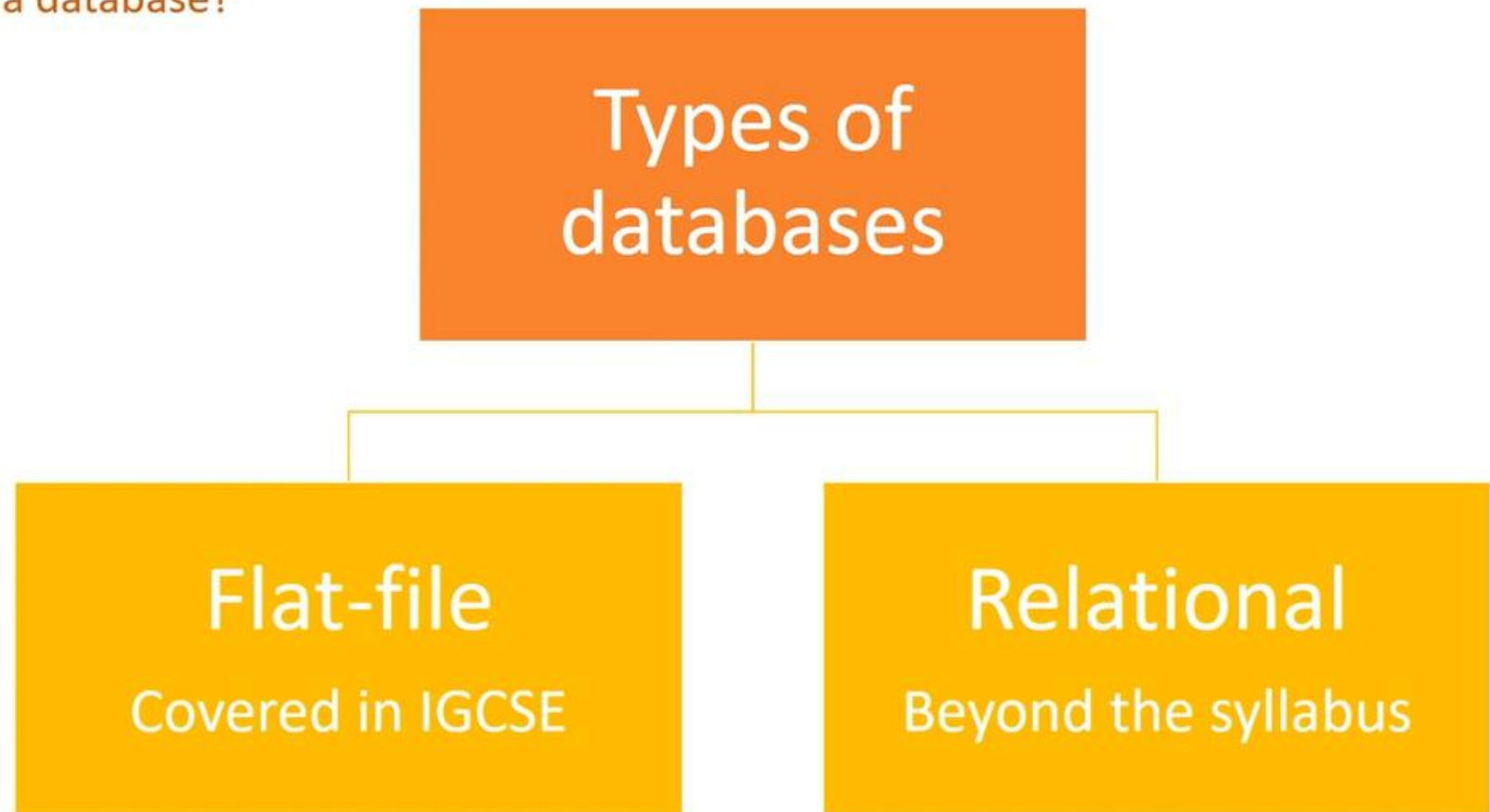
Databases are structured, persistent collections of data.

They allow us to arrange and store data in a series of tables for further querying and manipulation.



9.1 Define a **single-table database** from given **data storage** requirements

What is a database?



9.1 Define a **single-table database** from given **data storage** requirements

Flat-file database

A **flat-file database** consists of one or more **tables**. 

Each **table** consists of many separate **records** (rows). **Records** are identical in structure, though their contents will vary. 

Each **record** consists of a number of separate **fields** (columns). 

ClientsTbl

ID	Customer	Country	City	Date	Renew
1	Dongford International	USA	New York	13.09.2018	Y
2	Stateford Co.	France	Paris	14.11.2000	Y
3	Standard Ltd.	USA	San Francisco	21.03.2007	N
4	Nardis	Japan	Tokyo	30.05.2012	Y
5	Together Ltd.	USA	New York	01.04.1998	N
6	Nori	Japan	Tokyo	07.07.2005	Y
7	Intersec Co.	Germany	Berlin	15.11.2020	Y

9.1 Define a single-table database from given data storage requirements

Relational database

A **relational database** consists of multiple related **tables**.

In this example, changing the lecturer code for a course will update the associated field for all students on that course.

Students	Courses	Lecturers
Student_number	Course_code	Lecturer
1	A892	Alison Harper
10	E015	Austin O'Harriet
11	E210	Catherine Emmers
12	F451	Craig Sargent
13	F927	David Bunnell
14	G403	Julie James
2	H544	Jean Emmers
3	J564	James Halford
4	L900	Katherine Banks
5	M194	Lincoln Best
6	P101	Linda Cox
7	P202	Linda Lawton
8	P554	Matthew Purcell
9	Q947	Paul Smith
	S812	Simon Freelan



How separate database tables are linked together to create a relational database is beyond the scope of the IGCSE course. We have included it here for awareness, as it is something you are likely to come across should you continue to study computer science.

9.1 Define a single-table database from given data storage requirements

Databases

Each **field** in a database **table** has a defined **field type** – these include:

- Text/alphanumeric
- Character
- Boolean
- Integer
- Real
- Date/time

Integer	Text/alphanumeric	Text/alphanumeric	Text/alphanumeric	Date/time	Character
ID	Customer	Country	City	Date	Renew
1	Dongford International	USA	New York	13.09.2018	Y
2	Stateford Co.	France	Paris	14.11.2000	Y
3	Standard Ltd.	USA	San Francisco	21.03.2007	N
4	Nardis	Japan	Tokyo	30.05.2012	Y
5	Together Ltd.	USA	New York	01.04.1998	N

9.1 Define a single-table database from given data storage requirements

Primary keys

With a **database**, we should make sure each **record** within a **table** is uniquely identifiable.
Consequently, at least one **field** in each **record** should be guaranteed to be unique – this **field** is known as the **primary key**.



ID	Customer	Country	City	Date	Renew
1	Dongford International	USA	New York	13.09.2018	Y
2	Stateford Co.	France	Paris	14.11.2000	Y
3	Standard Ltd.	USA	San Francisco	21.03.2007	N
4	Nardis	Japan	Tokyo	30.05.2012	Y
5	Together Ltd.	USA	New York	01.04.1998	N
6	Nori	Japan	Tokyo	07.07.2005	Y
7	Intersec Co.	Germany	Berlin	15.11.2020	Y

9.1 Define a single-table database from given data storage requirements

Primary keys

While designing a **table**, you might discover one of the **fields** can act as a **primary key**.

In other situations, it will be necessary to create an additional **field** to act as the **primary key**.



Car_reg	Make	Model	Colour	Engine	Showroom
VK12 TKB	Vauxhall	Corsa	Red	1.2	Cars4U
K33 MAK	Audi	A1	Green	1.5	Cars4U
1975 CS	Fiat	Panda	Black	0.9	DriveIT
MI76 HEL	BMW	i8	Black	1.5	Need4Speed
BT32 UIK	Vauxhall	Carlton	Red	1.8	Cars4U
BL16 UDK	Mazda	Eunos 800	White	2.5	Need4Speed
GH63KDM	Fiat	Freemont	Green	1.2	DriveIT
ST26VUL	Fiat	Ducato	Blue	2.3	Cars4U
BP38 YTE	Audi	TT	Black	2.0	Need4Speed
W77 NDX	Audi	Avant	White	2.0	DriveIT
5690 AR	Vauxhall	Crossland	Red	1.2	DriveIT
MAG 247H	BMW	X5 (G05)	Blue	3.0	Need4Speed
NV27 XAW	Fiat	Toro	Black	2.4	Cars4U



Student ID	First name	Surname	Age	Tutor group
0001	Tony	Hilton	14	10C
0002	Amar	Aggarwal	15	10E
0003	Michael	Andrews	14	10C
0004	Claire	Smith	15	10E
0005	Nalini	Chahal	15	10C
0006	Chetan	Talwar	14	10C
0007	Andrew	Watts	14	10C
0008	Khasi	Vohra	15	10B
0010	James	Watkins	15	10C
0011	Sarah	Matthews	15	10B
0012	Tom	Nibblet	15	10E
0013	Mohandas	Majumdar	14	10E
0014	Andrew	Kuruvilla	14	10B

9.1 Define a single-table database from given data storage requirements

Databases validation

When a **database table** structure is first created, it can be assigned various **validation rules**.

Validation rules can be assigned to any **field** of a **table** – these include:

- Length check
- Format check
- Range check
- Presence check
- Custom rule such as *Apprentice*, *Expert* or *Master*.



We discussed various validation methods in detail in the following video:

- **83. CAMBRIDGE IGCSE (0478-0984) 7 Validation**

9.1 Define a **single-table database** from given **data storage** requirements

Single-table databases

Databases are structured, persistent collections of data.

Databases come in two forms – **single-table, flat file databases** and relational databases. You are only required to know about **single-table databases**.



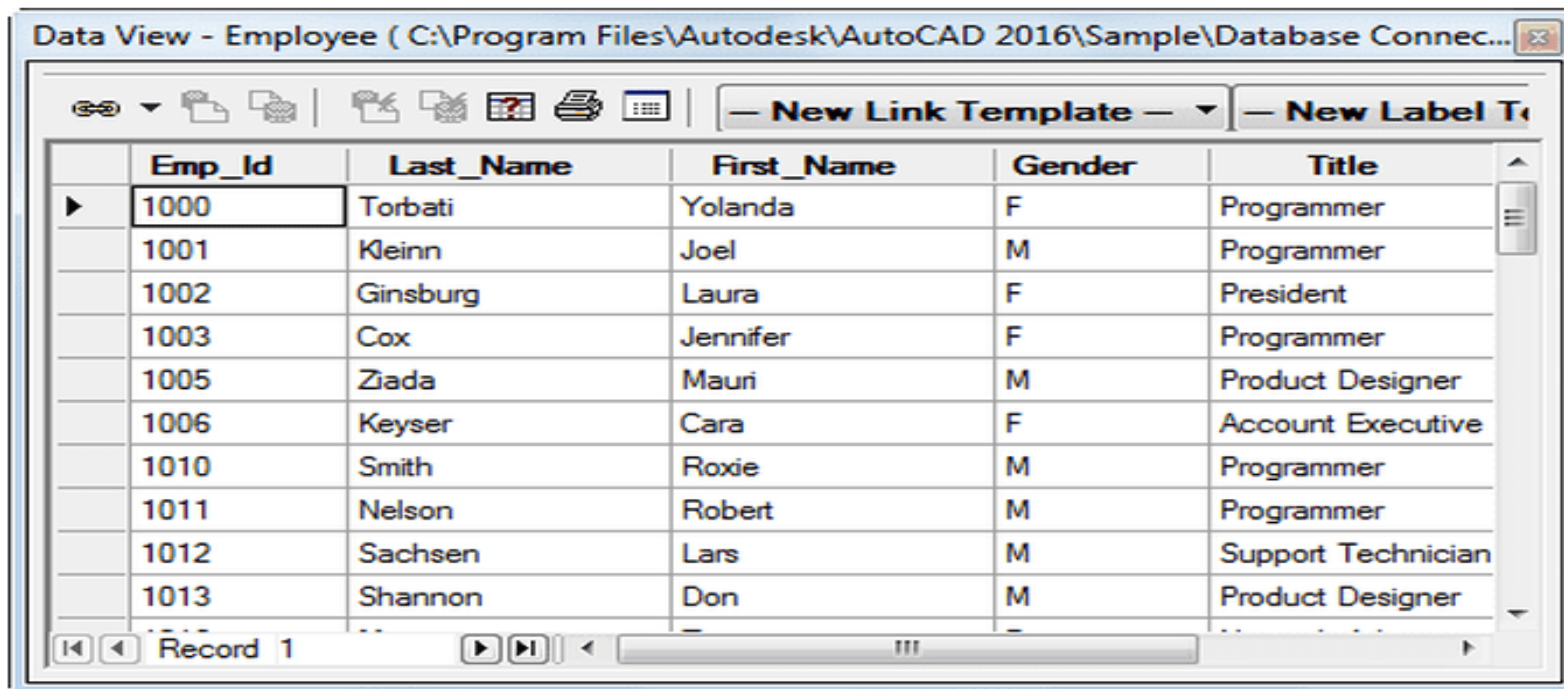
Flat file databases

- Consist of separate **tables**.
- Each **table** is made up of a number of **records**.
- Each **record** has a number of separate **fields**.
- **Fields** can be one of several **basic data types** – text/alphanumeric, character, Boolean, integer, real, date/time.
- Each **table** in a **database** should have a **primary key**. The **primary key field** must be a **field** that can uniquely identify each **record** in the **table**.
- **Database table fields** can be assigned various **validation rules** – length check, format check, range check, presence check, custom rule, etc.

9.1 Define a **single-table database** from given **data storage** requirements

A **single-table database** is a simple form of database where all the data is stored in a **single table**. Below, I'll define a single-table database based on given data storage requirements, including **fields**, **records**, and **validation**.

Here is an example database table, this table is storing employee data



The screenshot shows a window titled "Data View - Employee (C:\Program Files\Autodesk\AutoCAD 2016\Sample\Database Connec...". The window contains a table with 6 columns: Emp_Id, Last_Name, First_Name, Gender, and Title. The table has 12 rows of data. The first row is highlighted. The window also features a toolbar with various icons and a status bar at the bottom showing "Record 1".

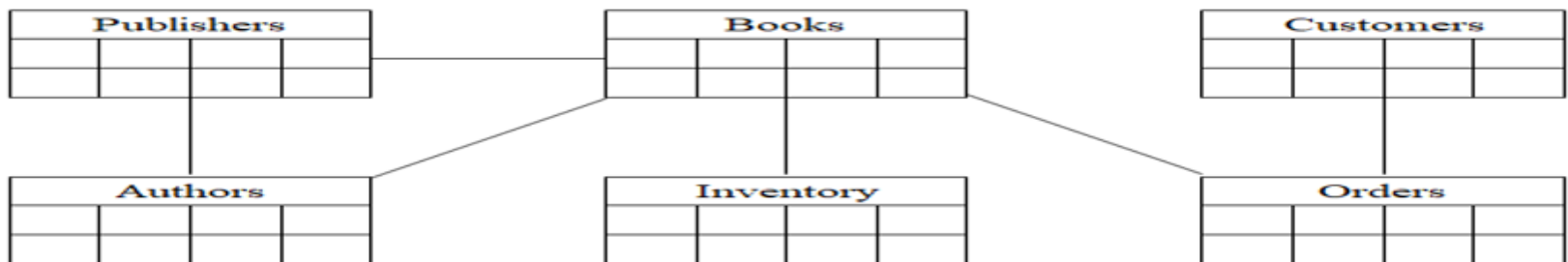
Emp_Id	Last_Name	First_Name	Gender	Title
1000	Torbati	Yolanda	F	Programmer
1001	Kleinn	Joel	M	Programmer
1002	Ginsburg	Laura	F	President
1003	Cox	Jennifer	F	Programmer
1005	Ziada	Mauri	M	Product Designer
1006	Keyser	Cara	F	Account Executive
1010	Smith	Roxie	M	Programmer
1011	Nelson	Robert	M	Programmer
1012	Sachsen	Lars	M	Support Technician
1013	Shannon	Don	M	Product Designer

9.1 Define a single-table database from given data storage requirements

Table

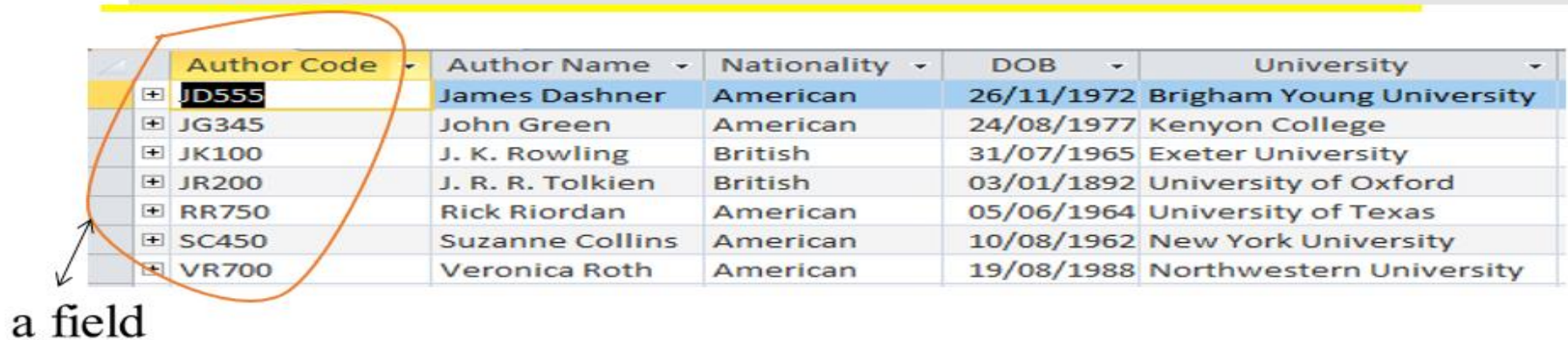
	Author Code ▾	Author Name ▾	Nationality ▾	DOB ▾	University ▾
+	JD555	James Dashner	American	26/11/1972	Brigham Young University
+	JG345	John Green	American	24/08/1977	Kenyon College
+	JK100	J. K. Rowling	British	31/07/1965	Exeter University
+	JR200	J. R. R. Tolkien	British	03/01/1892	University of Oxford
+	RR750	Rick Riordan	American	05/06/1964	University of Texas
+	SC450	Suzanne Collins	American	10/08/1962	New York University
+	VR700	Veronica Roth	American	19/08/1988	Northwestern University

A Database with Multiple Tables



9.1 Define a single-table database from given data storage requirements

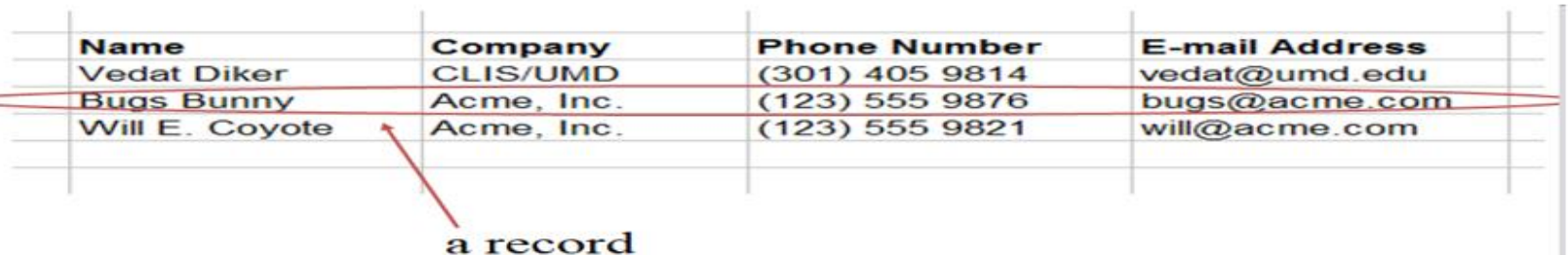
Field (Column)
below you have a total of 5 fields



Author Code	Author Name	Nationality	DOB	University
JD555	James Dashner	American	26/11/1972	Brigham Young University
JG345	John Green	American	24/08/1977	Kenyon College
JK100	J. K. Rowling	British	31/07/1965	Exeter University
JR200	J. R. R. Tolkien	British	03/01/1892	University of Oxford
RR750	Rick Riordan	American	05/06/1964	University of Texas
SC450	Suzanne Collins	American	10/08/1962	New York University
VR700	Veronica Roth	American	19/08/1988	Northwestern University

a field

Record (Row)
below you have total of 3 records



Name	Company	Phone Number	E-mail Address
Vedat Diker	CLIS/UMD	(301) 405 9814	vedat@umd.edu
Bugs Bunny	Acme, Inc.	(123) 555 9876	bugs@acme.com
Will E. Coyote	Acme, Inc.	(123) 555 9821	will@acme.com

a record

9.1 Define a **single-table database** from given **data storage** requirements

Validation: validation rules help ensure data integrity

When a **database table** structure is first created, it can be assigned various **validation rules**.

Validation rules can be assigned to any **field** of a **table** – these include:

- Length check
- Format check
- Range check
- Presence check

For Example:

- **Text/Alphanumeric:** Validate that FirstName and LastName are not empty and do not exceed a certain character limit.
- **Character (Gender):** Validate that Gender is either 'M' or 'F'.
- **Boolean (IsManager):** Ensure that IsManager is either **True** or **False**.
- **Integer (EmployeeID):** Ensure that EmployeeID is a positive integer.
- **Real (Salary):** Validate that Salary is a positive real number.
- **Date (BirthDate, JoinDate):** Validate that BirthDate and JoinDate are in a valid date format

9.1 Define a **single-table database** from given **data storage** requirements

1-Validation: validation rules help ensure data integrity

Validation checks are crucial in ensuring the accuracy, consistency, and integrity of input data in a system. They help identify and prevent errors or invalid data that could lead to incorrect results, system failures, or security breaches. Here are different types of validation checks along with their purposes.

Range Check: Verifies if the input value falls within an acceptable range.

Example: Checking if a user's age input falls within the range of 18 to 65

Length Check: Ensures that the length of the input data is within specified limits.

Example: Verifying if a password input has a length between 8 and 20 characters.

Type Check: Validates if the input data belongs to the expected data type.

Example: Checking if a user's input for a numerical value is an integer.

Presence Check: Ensures that a required field is not left empty.

Example: Verifying if a user's name input is not blank.

Format Check: Validates if the input data follows a specified format or pattern.

Example: Checking if an email address input follows the format "**example@example.com**".

Check Digit: Verifies the integrity of data, usually used in numerical data where a single digit is added to detect errors.

Example: Adding a check digit to a credit card number to detect input errors.

9.1 Define a **single-table database** from given **data storage** requirements

2- Verification:

Verification checks are essential for ensuring the accuracy and reliability of input data in a system. They help in confirming that the data entered is correct, valid, and consistent with the intended information. Here are different types of verification checks along with their purposes and examples:

Visual Check: Involves manually reviewing the input data to ensure it appears reasonable and accurate.

Example: A bank teller visually inspecting a customer's signature on a check to verify its authenticity.

Double Entry Check: Requires entering the same data twice independently and comparing the entries for consistency.

Example: When creating a new account, a user may be prompted to enter their email address twice to ensure they haven't made a typographical error

9.2 Suggest suitable basic **data types**

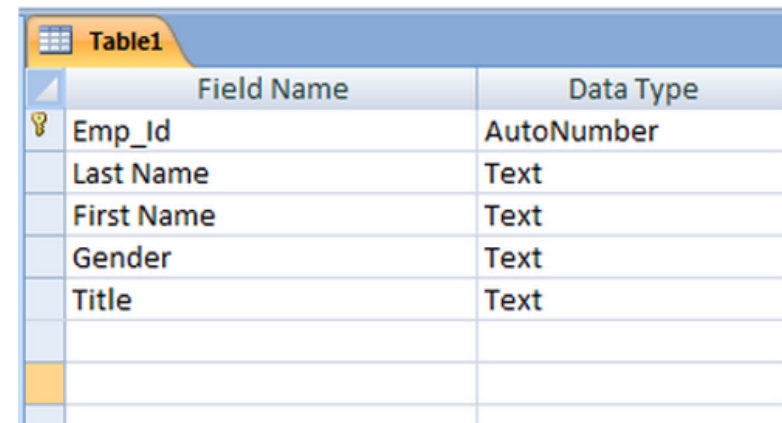
Data Types:

When setting up a database, each field must be given a name and a "Data type". If you are using Microsoft Access to set up your database, the process will look like this:

Computer program	Access
Integer	Number/Integer
Real	Number
Char	Text/field length 1
String	Text
Boolean	yes/no

Available data types:

- Alphanumeric (Text, Memo)
- Numeric (Number, Currency, etc.)
- Date/Time
- Boolean (Yes/No)



The screenshot shows a Microsoft Access table named 'Table1' in design view. The table has five fields: 'Emp_Id' (AutoNumber), 'Last Name' (Text), 'First Name' (Text), 'Gender' (Text), and 'Title' (Text). The 'Emp_Id' field is marked as the primary key with a key icon. The table structure is as follows:

	Field Name	Data Type
?	Emp_Id	AutoNumber
	Last Name	Text
	First Name	Text
	Gender	Text
	Title	Text

9.2 Suggest suitable basic **data types**

Available data types with description:

Data Type	Description
Text	Stores a short amount of text e.g a name, address, occupation
Integer	Stores a whole number e.g Year group = 10
Dates/ Times	Stores a date or time in a specified format e.g. 09/02/1988
Currency	Stores a monetary value with the currency symbol e.g \$44
Boolean (Yes / no)	Able to store one of two values e.g. yes or no, Male or Female
Real	Stores a number with a decimal point e.g. 5.6

9.3 Understand the purpose of a **primary key** and identify a suitable **primary key** for a given database table

The primary key in a relational database serves the fundamental purpose of uniquely identifying each record (row) in a table. It ensures that no two records can have the same key value, and it provides a way to establish relationships between tables. The primary key is crucial for maintaining data integrity and supporting efficient data retrieval.

Primary Key

Customers

Customer ID	Name	Company	Phone Number	E-mail Address
6273	Vedat Diker	CLIS/UMD	(301) 405 9814	vedat@umd.edu
3245	Bugs Bunny	Acme, Inc.	(123) 555 9876	bugs@acme.com
1324	Will E. Coyote	Acme, Inc.	(123) 555 9821	will@acme.com

primary key field

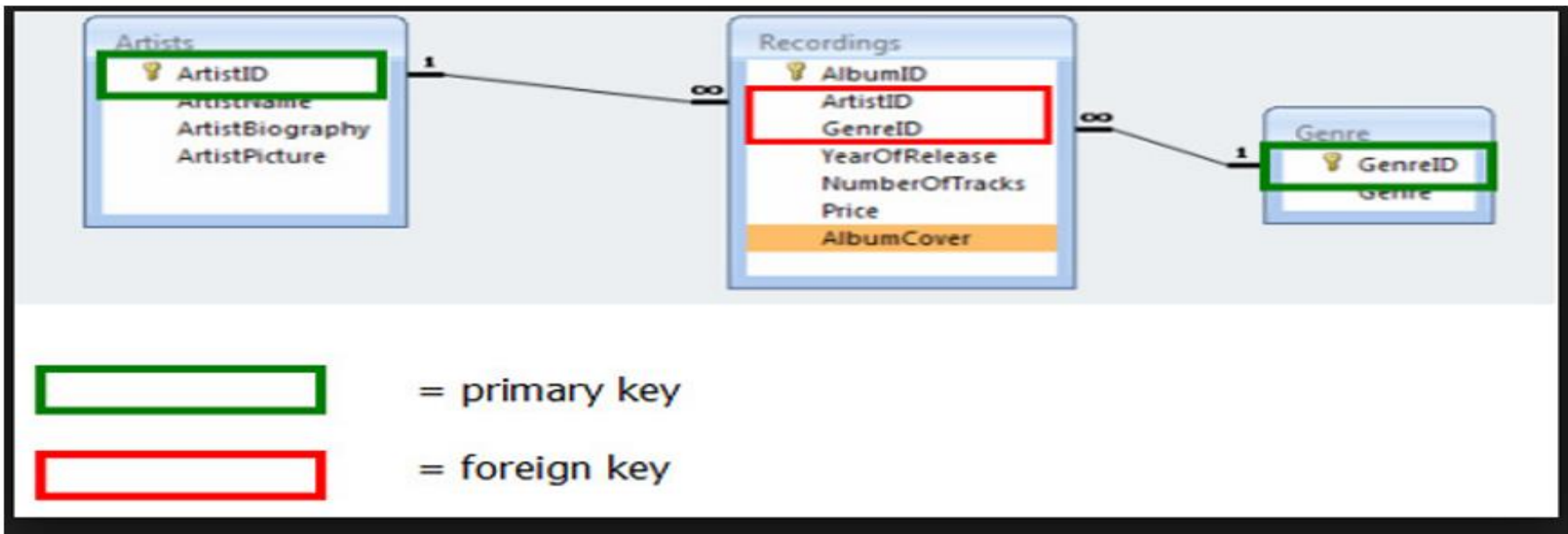
Primary key is a unique identifier of records in a table.

Primary key values may be generated manually or automatically.

9.3 Understand the purpose of a **primary key** and identify a suitable **primary key** for a given database table

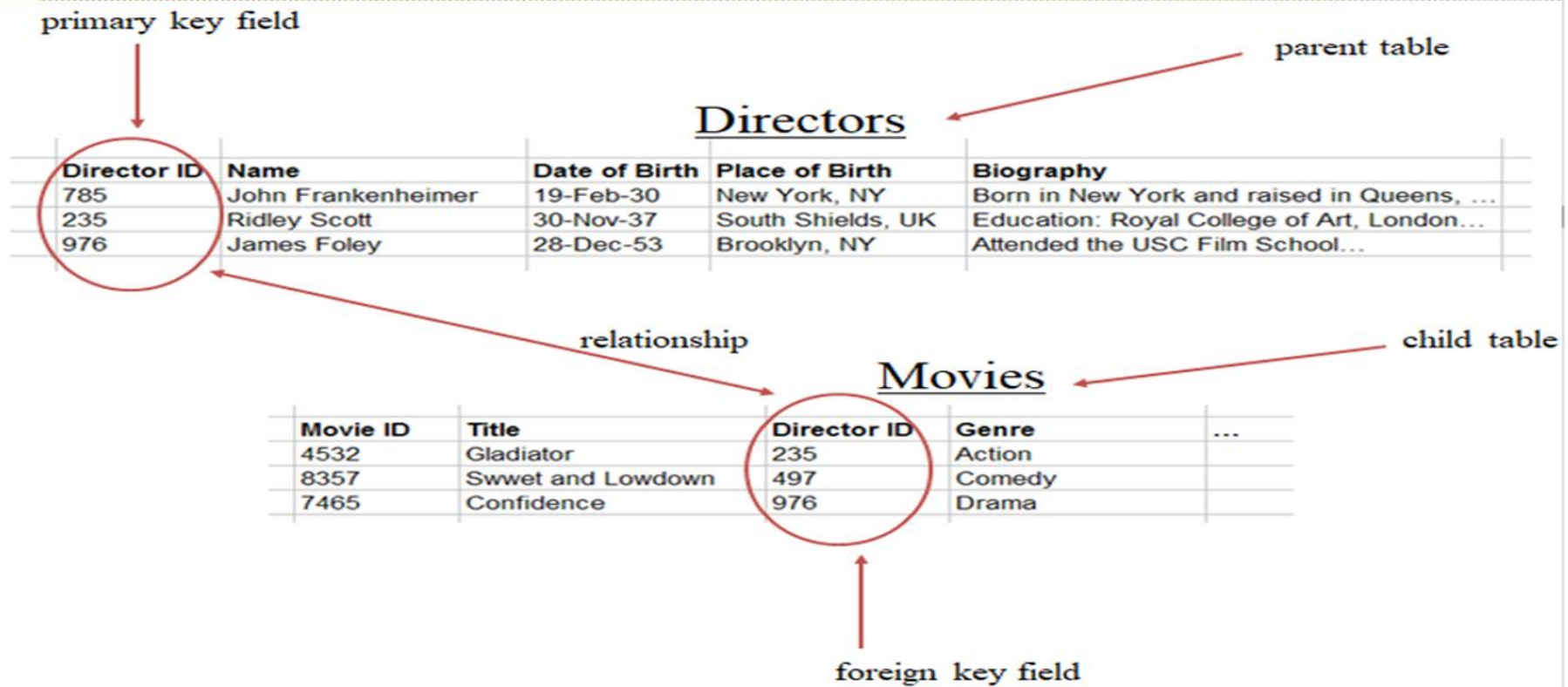
Foreign Key

A **foreign key** is a column or group of columns in a relational **database** table that provides a link between data in two tables. It acts as a cross-reference between tables because it references the primary **key** of another table, thereby establishing a link between them



9.3 Understand the purpose of a **primary key** and identify a suitable **primary key** for a given database table

Foreign Key



9.4 Read, understand and complete structured query language (SQL) scripts to query data stored in a single database table

Click the link below to watch the video for Database overview

<https://www.youtube.com/watch?v=uSjhgkCd6PM>

9 Databases

SQL

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6:43
ave.c

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

Information in a **database** is stored in **records**. Each **record** can contain a number of **fields**.

We can manipulate the data in these fields using a programming language called **Structured Query Language (SQL)**.

DB Browser for SQLite

File Edit View Help

Database Structure Browse Data Edit Pragma Execute SQL

Table: world

New Record Delete Record

	name	continent	area	population	gdp	capital
1	Afghanistan	Asia	652230	25500100	20364000000	Kabul
2	Albania	Europe	28748	2821977	12044000000	Tirana
3	Algeria	Africa	2381741	38700000	207021000000	Algiers
4	Andorra	Europe	468	76098	3222000000	Andorra la Vella
5	Angola	Africa	1246700	19183590	116308000000	Luanda
6	Antigua and B...	Caribbean	442	86295	1176000000	St. John's
7	Argentina	South America	2780400	42669500	477028000000	Buenos Aires
8	Armenia	Eurasia	29743	3017400	9950000000	Yerevan
9	Australia	Oceania	7692024	23545500	1564419000000	Canberra
10	Austria	Europe	83871	8504850	394458000000	Vienna
11	Azerbaijan	Asia	86600	9477100	68727000000	Baku
12	Bahamas	Caribbean		351461	8043000000	Nassau
13	Bahrain	Asia	765	1234571	30362000000	Manama

1 - 14 of 195

Go to: 1

UTF-8

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

Structured Query Language (SQL) statements supports a wide range of commands for manipulating a **database**.

The one you need to familiar with for the IGCSE exam is the **SELECT** statement.



9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

Structured Query Language (SQL) statements follow a strict format.

The format for the **SELECT** statement is as follows:

SELECT...

List the field(s) you want the SELECT query to display.

FROM...

List the table(s) the data will come from.

WHERE...

List the search criteria.

ORDER BY...

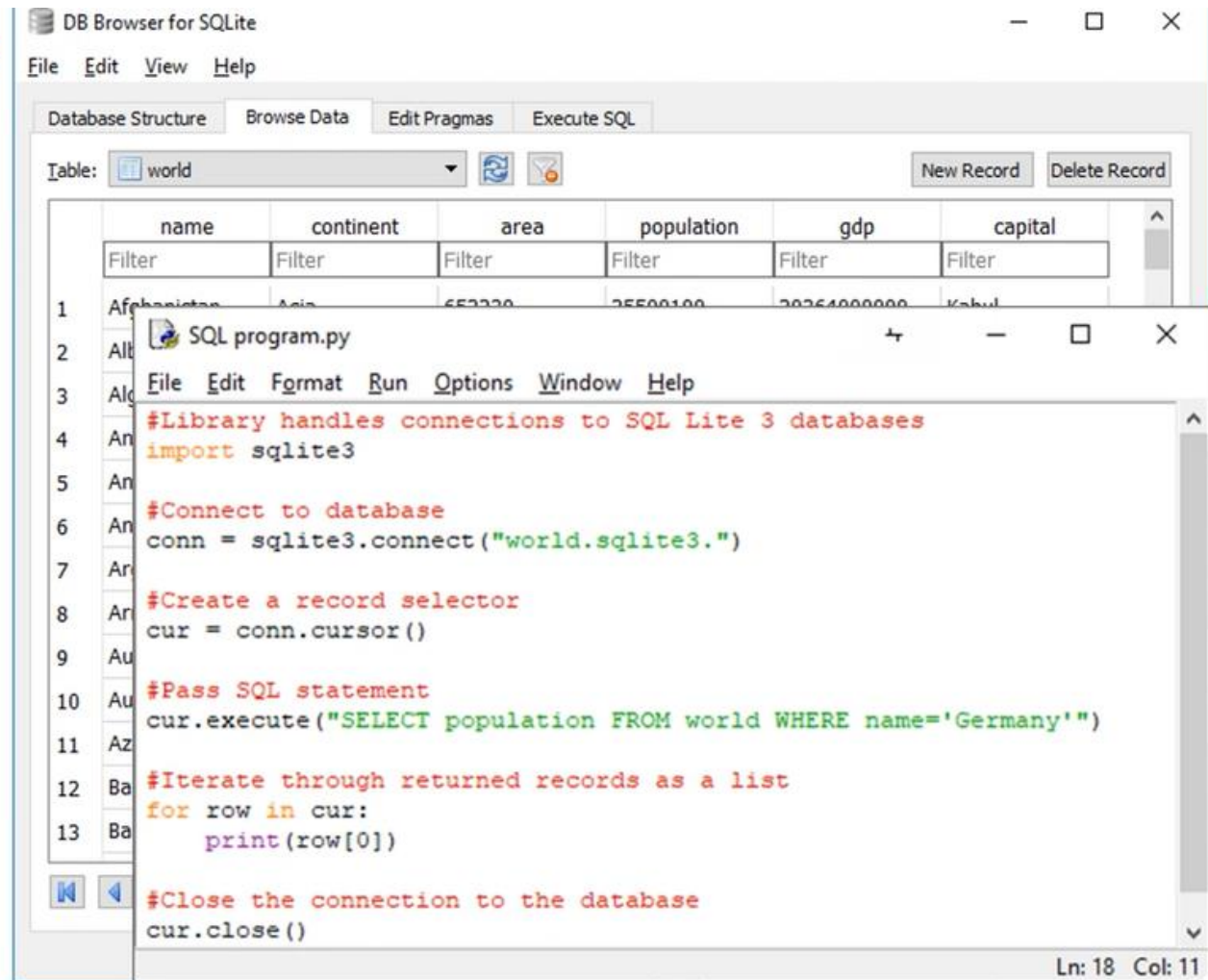
Optional criteria to sort the data in ascending (ASC) or descending (DESC) order.

9.4 Read, understand and complete structured query language (SQL) scripts to query data stored in a single database table

SQL: Structured Query Language

Here is a simple Python program that opens a connection to a **database** and finds **records** within it.

Most of the code in this program is beyond the IGCSE syllabus, so you don't need to worry about it at this stage.



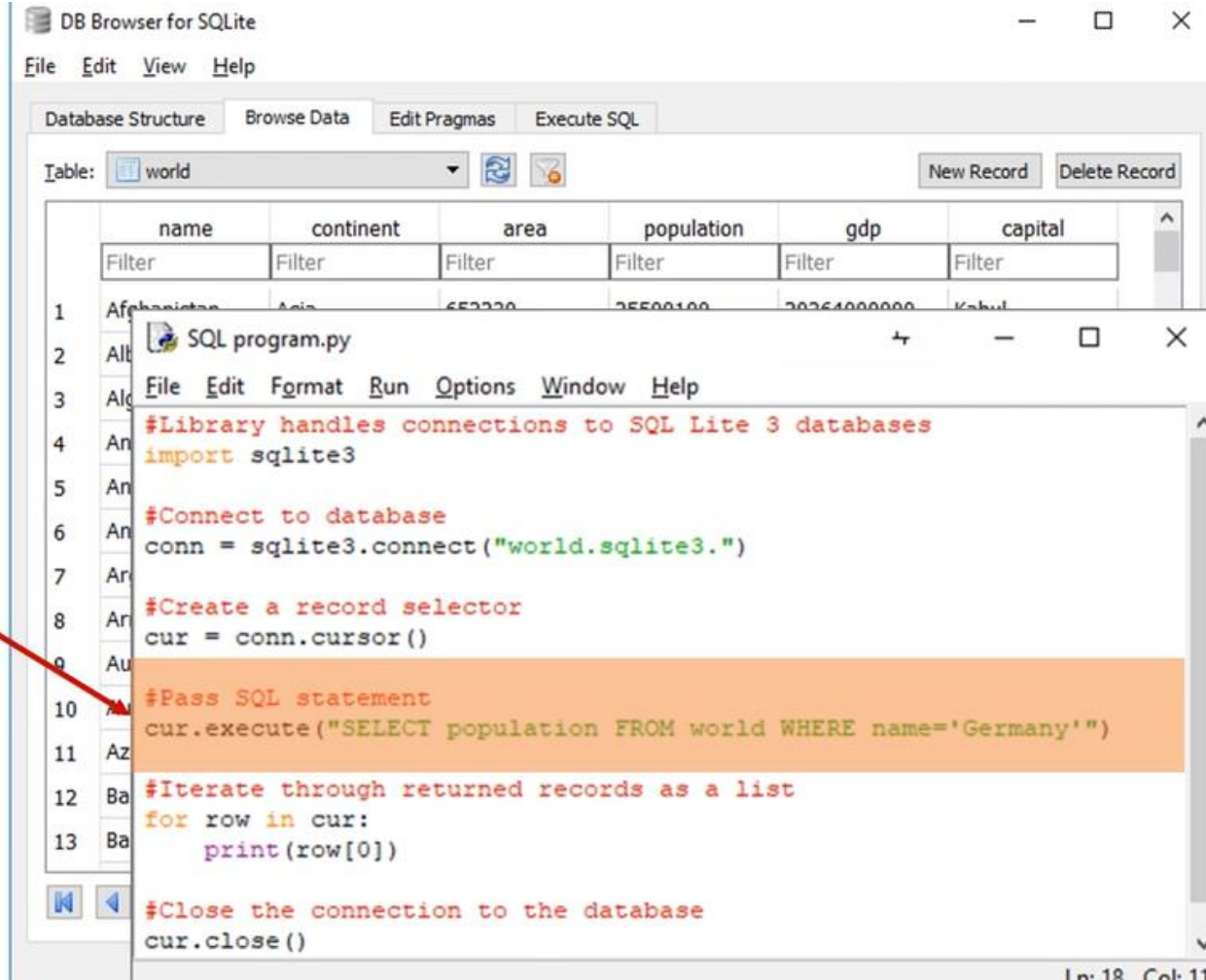
9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

The part of the program you *do* need to know about is highlighted here.

This is where we can issue **SQL** commands to the **database** for the purpose of finding and outputting data.

SELECT population
FROM world
WHERE name = "Germany"



The screenshot shows a Python IDE with a SQLite database browser window. The database browser window displays a table named 'world' with columns: name, continent, area, population, gdp, and capital. The Python script in the IDE is as follows:

```
SQL program.py
File Edit Format Run Options Window Help
#Library handles connections to SQL Lite 3 databases
import sqlite3

#Connect to database
conn = sqlite3.connect("world.sqlite3.")

#Create a record selector
cur = conn.cursor()

#Pass SQL statement
cur.execute("SELECT population FROM world WHERE name='Germany'")

#Iterate through returned records as a list
for row in cur:
    print(row[0])

#Close the connection to the database
cur.close()
```

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

```
SELECT population
FROM world
WHERE name = "Albania"
```

world

name	continent	area	population	gdp	capital
Afghanistan	Asia	652230	25500100	20364000000	Kabul
Albania	Europe	28748	2821977	12044000000	Tirana
Algeria	Africa	2381741	38700000	207021000000	Algiers
Andorra	Europe	468	76098	3222000000	Andorra la Vella
Angola	Africa	1246700	19183590	116308000000	Luanda
Antigua and B...	Caribbean	442	86295	1176000000	St. John's
Argentina	South America	2780400	42669500	477028000000	Buenos Aires
Armenia	Eurasia	29743	3017400	9950000000	Yerevan
Australia	Oceania	7692024	23545500	1564419000000	Canberra
Austria	Europe	83871	8504850	394458000000	Vienna
Azerbaijan	Asia	86600	9477100	68727000000	Baku
Bahamas	Caribbean		351461	8043000000	Nassau
Bahrain	Asia	765	1234571	30362000000	Manama

Output from this SQL query:

2821977

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

```
SELECT name, continent, area, population, gdp, capital
FROM world
WHERE name = "Algeria"
```

world

name	continent	area	population	gdp	capital
Afghanistan	Asia	652230	25500100	20364000000	Kabul
Albania	Europe	28748	2821977	12044000000	Tirana
Algeria	Africa	2381741	38700000	207021000000	Algiers
Andorra	Europe	468	76098	3222000000	Andorra la Vella
Angola	Africa	1246700	19183590	116308000000	Luanda
Antigua and B...	Caribbean	442	86295	1176000000	St. John's
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Azerbaijan	Asia	86600	9477100	68727000000	Baku
Bahamas	Caribbean		351461	8043000000	Nassau
Bahrain	Asia	765	1234571	30362000000	Manama

Output from this SQL query:

Algeria,Africa,238174,38700000,207021000000,Algiers

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

```
SELECT name, continent, area, population, gdp, capital
FROM world
WHERE name LIKE "A%"
AND population > 10000000
ORDER BY name ASC
```

world					
name	continent	area	population	gdp	capital
Afghanistan	Asia	652230	25500100	20364000000	Kabul
Albania	Europe	28748	2821977	12044000000	Tirana
Algeria	Africa	2381741	38700000	207021000000	Algiers
Andorra	Europe	468	76098	3222000000	Andorra la Vella
Angola	Africa	1246700	19183590	116308000000	Luanda
Antigua and B...	Caribbean	442	86295	1176000000	St. John's
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Armenia	Eurasia	29743	3017400	9950000000	Yerevan
Australia	Oceania	7692024	23545500	1564419000000	Canberra
Austria	Europe	83871	8504850	394458000000	Vienna
Azerbaijan	Asia	86600	9477100	68727000000	Baku
Bahamas	Caribbean		351461	8043000000	Nassau
Bahrain	Asia	765	1234571	30362000000	Manama

Output from this SQL query:

```
Afghanistan,Asia,652230,25500110,20364000000,Kabul
Algeria,Africa,238174,38700000,207021000000,Algiers
Angola,Africa,1246700,19183590,116308000000,Luanda
Argentina,South America,2780400,42669500,477028000000,Buenos Aires
Australia,Oceania,7692024,2354550,1564419000000,Canberra
```

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

```
SELECT COUNT(*)  
FROM world  
WHERE continent = "Asia"
```

world					
name	continent	area	population	gdp	capital
Afghanistan	Asia	652230	25500100	20364000000	Kabul
Albania	Europe	28748	2821977	12044000000	Tirana
Algeria	Africa	2381741	38700000	207021000000	Algiers
Andorra	Europe	468	76098	3222000000	Andorra la Vella
Angola	Africa	1246700	19183590	116308000000	Luanda
Antigua and B...	Caribbean	442	86295	1176000000	St. John's
Argentina	South America	2780400	42669500	477028000000	Buenos Aires
Armenia	Eurasia	29743	3017400	9950000000	Yerevan
Australia	Oceania	7692024	23545500	1564419000000	Canberra
Austria	Europe	83871	8504850	394458000000	Vienna
Azerbaijan	Asia	86600	9477100	68727000000	Baku
Bahamas	Caribbean		351461	8043000000	Nassau
Bahrain	Asia	765	1234571	30362000000	Manama

Output from this SQL query:

3

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

SQL: Structured Query Language

```
SELECT SUM(population)
FROM world
WHERE continent = "Europe"
```

world					
name	continent	area	population	gdp	capital
Afghanistan	Asia	652230	25500100	20364000000	Kabul
Albania	Europe	28748	2821977	12044000000	Tirana
Algeria	Africa	2381741	38700000	207021000000	Algiers
Andorra	Europe	468	76098	3222000000	Andorra la Vella
Angola	Africa	1246700	19183590	116308000000	Luanda
Antigua and B...	Caribbean	442	86295	1176000000	St. John's
Argentina	South America	2780400	42669500	477028000000	Buenos Aires
Armenia	Eurasia	29743	3017400	9950000000	Yerevan
Australia	Oceania	7692024	23545500	1564419000000	Canberra
Austria	Europe	83871	8504850	394458000000	Vienna
Azerbaijan	Asia	86600	9477100	68727000000	Baku
Bahamas	Caribbean		351461	8043000000	Nassau
Bahrain	Asia	765	1234571	30362000000	Manama

Output from this SQL query:

11402655

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table



SQL: Structured Query Language

SQL is used to create, delete, modify and manipulate records in a database. Basic commands include:

- SELECT which fields to be returned.
- FROM which table – databases can have more than one table, each with their own unique name.
- WHERE records meet a condition.
- LIKE can be used as a wildcard.
- ORDER BY: Choose how the data is organised when it is retrieved (e.g., ORDER BY name).
 - ORDER BY <field> ASC: Order by the selected field in ascending order.
 - ORDER BY <field> DESC: Order by the selected field in descending order.
- SELECT SUM <field>: Specifies fields for calculation – should be integer or real.
- SELECT COUNT <field>: Counts fields that meet the specified criteria.

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Click the link below to access the SQL Compiler:
<https://www.programiz.com/sql/online-compiler/>

1- Basic **SELECT** statement:

// Select all columns from the Customers table

SELECT * **FROM** Customers;

Customers

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA
3	David	Robinson	22	UK
4	John	Reinhardt	25	UK
5	Betty	Doe	28	UAE

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

2- **SELECT specific columns**

// Select specific columns from the Customers table

SELECT customer_id, first_name, last_name, age
FROM Customers;

customer_id	first_name	last_name	age
1	John	Doe	31
2	Robert	Luna	22
3	David	Robinson	22
4	John	Reinhardt	25
5	Betty	Doe	28

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

3- Filtering rows using **WHERE**

// Select customers from the USA

```
SELECT *  
FROM Customers  
WHERE country = 'USA';
```

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
2	Robert	Luna	22	USA

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

4- Ordering results using **ORDER BY**

// Select customers and order them by age in descending order

```
SELECT *  
FROM Customers  
ORDER BY age DESC;
```

customer_id	first_name	last_name	age	country
1	John	Doe	31	USA
5	Betty	Doe	28	UAE
4	John	Reinhardt	25	UK
2	Robert	Luna	22	USA
3	David	Robinson	22	UK

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Example 1: Create TABLE Students

Create the Students Table: We begin by defining the structure of our table. In SQL, this is done using the CREATE TABLE statement. We specify the names and data types of each field.

```
CREATE TABLE Students (  
    StudentID INT AUTO_INCREMENT PRIMARY KEY,  
    Name VARCHAR(50),  
    Age INT,  
    Gender VARCHAR(10),  
    Grade INT  
);
```

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Example 1- INSERT Students data

Insert Student Data:

INSERT INTO Students (StudentID, Name, Age, Gender, Grade) **VALUES**

(1, 'Nuriya', 15, 'Female', 95),
(2, 'Artur', 15, 'Male', 68),
(3, 'Danial', 15, 'Male', 70),
(4, 'Inaaya', 15, 'Female', 95),
(5, 'Harsh', 15, 'Male', 90),
(6, 'Ivan', 15, 'Male', 80),
(7, 'Eker', 15, 'Male', 65),
(8, 'Beiza', 15, 'Female', 81),
(9, 'Nicholas', 15, 'Male', 66),
(10, 'Darius', 15, 'Male', 62),
(11, 'Samina', 15, 'Female', 90),
(12, 'Emir', 15, 'Male', 95);

Students

StudentID	Name	Age	Gender	Grade
1	Nuriya	15	Female	95
2	Artur	15	Male	68
3	Danial	15	Male	70
4	Inaaya	15	Female	95
5	Harsh	15	Male	90
6	Ivan	15	Male	80
7	Eker	15	Male	65
8	Beiza	15	Female	81
9	Nicholas	15	Male	66
10	Darius	15	Male	62
11	Samina	15	Female	90
12	Emir	15	Male	95

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Example 1:

1: Retrieve names and ages of female students who scored above 90 in exams, ordered by age.

```
SELECT Name, Age
FROM Students
WHERE Gender = 'Female' AND Grade > 90
ORDER BY Age;
```

Output

Name	Age
Nuriya	15
Inaaya	15

2): Calculate the total number of male students in the database.

```
SELECT COUNT(*)
FROM Students
WHERE Gender = 'Male';
```

Output

COUNT(*)
8

3): Calculate the total sum of grades for all students.

```
SELECT SUM(Grade)
FROM Students;
```

Output

SUM(Grade)
957

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Example 2- **CREATE TABLE Sales**

```
CREATE TABLE Sales (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(255),  
    Category VARCHAR(50),  
    QuantitySold INT,  
    UnitPrice DECIMAL(10, 2)  
);
```

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Example 2: INSERT INTO TABLE Sales

// Inserting 5 records into the Sales table

INSERT INTO Sales (ProductID, ProductName, Category, QuantitySold, UnitPrice)
VALUES

(1, 'Laptop', 'Electronics', 10, 800.00),
(2, 'Smartphone', 'Electronics', 20, 500.00),
(3, 'Desk Chair', 'Furniture', 5, 120.00),
(4, 'LED TV', 'Electronics', 15, 1000.00),
(5, 'Coffee Table', 'Furniture', 8, 150.00);

Sales

ProductID	ProductName	Category	QuantitySold	UnitPrice
1	Laptop	Electronics	10	800
2	Smartphone	Electronics	20	500
3	Desk Chair	Furniture	5	120
4	LED TV	Electronics	15	1000
5	Coffee Table	Furniture	8	150

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Example 2: - Aggregation functions - **SUM** and **COUNT**

// Calculate the total quantity sold and the number of sales

```
SELECT SUM(QuantitySold) AS TotalQuantity, COUNT(*) AS QuantitySold  
FROM Sales;
```

TotalQuantity	QuantitySold
58	5

9.4 Read, understand and complete **structured query language (SQL)** scripts to query data stored in a single database table

Example 2- Combining conditions with **WHERE** and using aggregate functions

// Calculate the total quantity sold to customers from the UK

```
SELECT country, SUM(QuantitySold) AS TotalQuantity
FROM Sales
JOIN Customers ON Sales.ProductID = Customers.customer_id
WHERE country = 'UK'
GROUP BY country;
```

country	TotalQuantity
UK	20

// Or you can also write your SQL like this below and still will give you the same result:

```
SELECT Customers.country, SUM(Sales.QuantitySold) AS TotalQuantity
FROM Sales
JOIN Customers ON Sales.ProductID = Customers.customer_id
WHERE Customers.country = 'UK'
GROUP BY Customers.country;
```

country	TotalQuantity
UK	20

9.1 Database

Past Paper 2 Exams Questions and Answers:

Question 1:

A database table called **TVRange** shows the main features and prices of a range of televisions.

TVCode	ScreenSize	Satellite	SmartTV	SoundBar	Price\$
TV90SaSmSd	90	YES	YES	YES	9750.00
TV75SaSmSd	75	YES	YES	YES	8500.00
TV75SaSd	75	YES	NO	YES	8000.00
TV65SaSmSd	65	YES	YES	YES	6000.00
TV65SmSd	65	NO	YES	YES	5000.00
TV65SaSd	65	YES	NO	YES	5000.00
TV55SaSmSd	55	YES	YES	YES	4000.00
TV55SaSd	55	YES	NO	YES	3500.00
TV55SmSd	55	NO	YES	YES	3500.00
TV50SaSmSd	50	YES	YES	YES	2500.00
TV50Sa	50	YES	NO	NO	1750.00
TV50Sm	50	NO	YES	NO	1750.00
TV40Sa	40	YES	NO	NO	1200.00
TV40	40	NO	NO	NO	950.00
TV32	32	NO	NO	NO	650.00

(a) Give the name of the field that is most suitable to be the primary key.

State the reason for this choice.

Field

Reason

.....

9.1 Database

Question 1:

- (b) The database uses the data types:
- text
 - character
 - Boolean
 - integer
 - real
 - date/time.

Complete the table to show the most appropriate data type for each field.
Each data type must be different.

Field	Data type
TVCode	
ScreenSize	
SmartTV	
Price\$	

[2]

- (c) Complete the structured query language (SQL) query to return the television (TV) code, screen size and price of all Smart TVs in the database table.

```
SELECT TVCode, ..... , .....  
..... TVRange  
  
WHERE SmartTV = .....
```

[4]

9.1 Database

Answer 1:

Question	Answer	Marks										
10(a)	One mark for the correct field name One mark for the correct reason For example: TVCode Each entry in this field is a unique identifier	2										
10(b)	Two marks for four correct answers. One mark for two or three correct answers. <table><tr><th>Field</th><th>Data type</th></tr><tr><td>TVCode</td><td>Text</td></tr><tr><td>ScreenSize</td><td>Integer</td></tr><tr><td>SmartTV</td><td>Boolean</td></tr><tr><td>Price\$</td><td>Real</td></tr></table>	Field	Data type	TVCode	Text	ScreenSize	Integer	SmartTV	Boolean	Price\$	Real	
Field	Data type											
TVCode	Text											
ScreenSize	Integer											
SmartTV	Boolean											
Price\$	Real											
10(c)	One mark for each correct answer ScreenSize Price\$ FROM YES Correct code: <pre>SELECT TVCode, ScreenSize, Price\$ FROM TVRange WHERE SmartTV = YES;</pre>											

9.1 Database

Question 2:

A music streaming service has a new database table named `Songs` to store details of songs available for streaming. The table contains the fields:

- `SongNumber` – the catalogue number, for example AG123
- `Title` – the title of the song
- `Author` – the name of the song writer(s)
- `Singer` – the name of the singer(s)
- `Genre` – the type of music, for example rock
- `Minutes` – the length of the song in minutes, for example 3.75
- `Recorded` – the date the song was recorded.

(a) Identify the field that will be the most appropriate primary key for this table.

..... [1]

(b) Complete the table to identify the most appropriate data type for the fields in `Songs`

Field	Data type
<code>SongNumber</code>	
<code>Title</code>	
<code>Recorded</code>	
<code>Minutes</code>	

[2]

9.1 Database

Question 2:

(c) Explain the purpose of the structured query language (SQL) statements.

```
SUM (Minutes) FROM Songs WHERE Genre = "rock";
```

```
COUNT (Title) FROM Songs WHERE Genre = "rock";
```

.....

.....

.....

.....

.....

..... [3]

9.1 Database

Answer 2:

PUBLISHED

Question	Answer	Marks										
10(a)	SongNumber	1										
10(b)	One mark for every two correct data types <table><tr><th>Field</th><th>Data Type</th></tr><tr><td>SongNumber</td><td>Text/Alphanumeric</td></tr><tr><td>Title</td><td>Text/Alphanumeric</td></tr><tr><td>Recorded</td><td>Date/time</td></tr><tr><td>Minutes</td><td>Real</td></tr></table>	Field	Data Type	SongNumber	Text/Alphanumeric	Title	Text/Alphanumeric	Recorded	Date/time	Minutes	Real	2
Field	Data Type											
SongNumber	Text/Alphanumeric											
Title	Text/Alphanumeric											
Recorded	Date/time											
Minutes	Real											
10(c)	One mark for each point - to find the total number of minutes of music - to find the total number of songs - available for the genre rock	3										

9.1 Database

Question 3:

A database table called `Site1` stores details of some holiday homes at a holiday park. The database shows the type of home, number of guests, whether it is privately owned and the weekly rate to hire it.

Name	Type	Private	Rate\$	NumberGuest
Bay Lodge	Lodge	NO	1000	10
Bay View	Cabin	NO	400	4
Blue Skies	Cabin	NO	350	4
Cliff View	Cabin	NO	650	6
Coppice Lodge	Lodge	NO	1200	12
Green Lodge	Lodge	NO	1000	8
Henry	Cabin	YES	300	2
Hikers' Rest	Retreat	NO	750	6
Poppy	Cabin	NO	300	2
Summer Joy	Retreat	YES	750	6
Valley View	Cabin	NO	600	6
West Lodge	Lodge	YES	1200	12

(a) State the number of fields and the number of records in this database table.

Fields

Records

9.1 Database

Question 3:

(b) Describe the purpose of a primary key.

.....

..... [1]

(c) The database uses the data types:

- alphanumeric
- character
- Boolean
- integer
- real
- date/time.

Complete the table to show the most appropriate data type for each field.

Field	Data type
Type	
Private	
Rate\$	
NumberGuest	

9.1 Database

Question 3:

(d) Give the output that would be produced by the structured query language (SQL) statement:

```
SELECT Name, NumberGuest, Rate$  
FROM Site1  
WHERE NumberGuest >= 10;
```

.....

.....

.....

.....

.....

..... [3]

9.1 Database

Answer 3:

10(a)	One mark for each correct answer Fields 5 Records 12	2										
10(b)	to uniquely identify a record	1										
10(c)	Two marks for four correct answers. One mark for two or three correct answers. <table border="1"><thead><tr><th>Field</th><th>Data type</th></tr></thead><tbody><tr><td>Type</td><td>Alphanumeric</td></tr><tr><td>Private</td><td>Boolean</td></tr><tr><td>Rate\$</td><td>Integer</td></tr><tr><td>NumberGuest</td><td>Integer</td></tr></tbody></table>	Field	Data type	Type	Alphanumeric	Private	Boolean	Rate\$	Integer	NumberGuest	Integer	2
Field	Data type											
Type	Alphanumeric											
Private	Boolean											
Rate\$	Integer											
NumberGuest	Integer											
10(d)	One mark per mark point, max three • data correctly extracted in any two rows • data correctly extracted in third row • data in correct order horizontally and vertically Example answer: Bay Lodge 10 1000 Coppice Lodge 12 1200 West Lodge 12 1200	3										

- 9 A shop that sells books has set up a new database table called `BookList` to store book details. Part of this table is given.

Question 4:

CatNo	Title	Fiction	Author	PaperBack	Price	StockLevel
BK01	The Princes' Story	Yes	B Penn	Yes	4.50	500
BK02	The Princesses' Story	Yes	B Penn	Yes	4.50	350
BK03	Computer Science	No	Way Yu	Yes	19.99	20
BK04	The Modern World	No	P Patel	No	25.00	5
BK05	The Ancient World	Yes	P Patel	No	25.00	5
BK06	Computer Science	No	R Dale	Yes	27.35	8
BK07	The Princes' Story	Yes	B Penn	No	12.50	3
BK08	The Princesses' Story	Yes	B Penn	No	12.50	0
BK12	Famous Five	Yes	E Bly	Yes	2.75	45
BK15	Secret Seven	Yes	E Bly	Yes	2.75	25
BK16	The Last Knight	Yes	P Mann	Yes	5.99	7
BK17	The Dark Tower	Yes	P Mann	Yes	5.99	5
BK19	The Final Chase	Yes	P Mann	Yes	5.99	5
BK21	Maths Today Part 1	No	B Ward	Yes	6.75	25
BK22	Maths Today Part 2	No	B Ward	Yes	6.75	15
BK23	Maths Today Part 3	No	B Ward	Yes	6.75	10
BK26	Maths Today Workbook	No	B Ward	Yes	6.75	30
BK27	Knitting for Beginners	No	A Smith	Yes	6.99	3
BK30	Woodwork for Beginners	No	A Smith	Yes	6.99	4
BK31	Networking for Beginners	No	A Smith	Yes	6.99	0

- (a) State the number of records in this part of the database table.

..... [1]

- (b) (i) Give the name of the field that would be used for the primary key.

..... [1]

- (ii) State the reason for choosing this field for the primary key.

.....
..... [1]

- (c) Complete the table to identify the most appropriate data type for each field based on the data shown in the table `BookList`

Field	Data type
CatNo	
Title	
Fiction	
Price	

Question 4:

[2]

- (d) Write the output from this structured query language (SQL) statement.

```
SELECT CatNo, Title, Author
FROM BookList
WHERE StockLevel = 0;
```

.....

.....

.....

..... [2]

- (e) Complete this SQL statement to display all the titles by the author B Penn.

```
SELECT .....
FROM .....
WHERE .....;
```

[2]

9.1 Database

Answer 4:

Question	Answer	Marks										
9(a)	20	1										
9(b)(i)	CatNo	1										
9(b)(ii)	It is a unique identifier // no repeated values	1										
9(c)	Two marks for 4 correct data types or one mark for 2 or 3 correct data types <table><tr><th>Field</th><th>Data type</th></tr><tr><td>CatNo</td><td>Text/Alphanumeric</td></tr><tr><td>Title</td><td>Text/alphanumeric</td></tr><tr><td>Fiction</td><td>Boolean</td></tr><tr><td>Price</td><td>Real</td></tr></table>	Field	Data type	CatNo	Text/Alphanumeric	Title	Text/alphanumeric	Fiction	Boolean	Price	Real	2
Field	Data type											
CatNo	Text/Alphanumeric											
Title	Text/alphanumeric											
Fiction	Boolean											
Price	Real											
9(d)	One mark for each correct row <table><tr><td>BK08</td><td>The Princesses' Story</td><td>B Penn</td></tr><tr><td>BK31</td><td>Networking for Beginners</td><td>A Smith</td></tr></table>	BK08	The Princesses' Story	B Penn	BK31	Networking for Beginners	A Smith	2				
BK08	The Princesses' Story	B Penn										
BK31	Networking for Beginners	A Smith										
9(e)	One mark if two correct or two marks if completely correct Title BookList Author = "B Penn" // Author = 'B Penn' // Author Like "B Penn"	2										

9.1 Database

Question 5 :

A database table, 2018MOV, is used to keep a record of movie details.

CatNo	Title	Genre1	Genre2	Blu-ray	DVD	Streaming
18m01	Power Rangers	Adventure	Fantasy	Yes	No	Yes
18m02	Baywatch	Comedy	Drama	Yes	No	Yes
18m03	Table 19	Comedy	Drama	Yes	Yes	No
18m04	Wonder Woman	Action	Fantasy	Yes	No	Yes
18m05	Justice League	Action	Fantasy	Yes	Yes	Yes
18m06	Twilight	Thriller	Action	Yes	Yes	No
18m07	Ant Man	Action	Fantasy	No	Yes	No
18m08	Venice Beach	Action	History	No	Yes	No
18m12	Fast Five	Action	Thriller	No	Yes	No
18m15	King Kong	Adventure	Fantasy	No	Yes	No
18m16	Transformers: The Last Knight	Action	Sci-Fi	Yes	Yes	Yes
18m17	The Dark Tower	Fantasy	Sci-Fi	Yes	Yes	No
18m19	Beauty and the Beast	Fantasy	Romance	Yes	Yes	Yes
18m21	The Mummy	Action	Fantasy	No	No	Yes
18m22	Star Wars: Episode VIII	Sci-Fi	Action	Yes	No	Yes
18m23	Guardians of the Galaxy	Action	Sci-Fi	Yes	Yes	Yes
18m26	Thor	Action	Sci-Fi	No	Yes	Yes
18m27	Twilight	Fantasy	Sci-Fi	No	No	Yes
18m30	Beneath	Action	Fantasy	Yes	No	No
18m31	Despicable Me	Animation	Action	Yes	Yes	No

- (a) State the number of records in the database table.
..... [1]
- (b) (i) Give the name of the field that would be used for the primary key.
..... [1]
- (ii) State the reason for choosing this field for the primary key.
.....
..... [1]

9.1 Database

Question 5 :

(c) Complete the table to identify the most appropriate data type for each field based on the data shown in the database table, 2018MOV.

Field	Data type
CatNo	
Title	
Genrel	
Streaming	

[2]

(d) Complete the structured query language (SQL) to return the category number and title for all Comedy movies.

```
SELECT CatNo, Title
..... 2018MOV
WHERE Genrel = .....
```

[2]

9.1 Database

Answer 5 :

Question	Answer	Marks										
11(a)	20	1										
11(b)(i)	CatNo	1										
11(b)(ii)	it is a unique identifier	1										
11(c)	Two marks for four correct answers. One mark for two or three correct answers. <table><tr><th>Field</th><th>Data type</th></tr><tr><td>CatNo</td><td>Text</td></tr><tr><td>Title</td><td>Text</td></tr><tr><td>Genre1</td><td>Text</td></tr><tr><td>Streaming</td><td>Boolean / Text</td></tr></table>	Field	Data type	CatNo	Text	Title	Text	Genre1	Text	Streaming	Boolean / Text	2
Field	Data type											
CatNo	Text											
Title	Text											
Genre1	Text											
Streaming	Boolean / Text											
11(d)	FROM "Comedy"	2										

9.1 Database

Question 6 :

A database, PROPERTY, was set up to show the prices of properties for sale and the features of each property. Part of the database is shown below.

Property Type	Brochure No	Number of Bedrooms	Number of Bathrooms	Garden	Garage	Price in \$
Bungalow	B17	7	4	Yes	Yes	750,000
Apartment	A09	2	1	No	No	100,000
House	H10	4	2	Yes	No	450,000
House	H13	3	2	Yes	No	399,000
Apartment	A01	2	2	No	Yes	95,000
Apartment	A16	1	1	No	No	150,000
House	H23	3	1	No	Yes	250,000
House	H46	2	1	Yes	Yes	175,000

(a) Give the number of fields that are in each record.

.....[1]

9.1 Database

Question 6 :

(b) State which field you would choose for the primary key.

.....

Give a reason for choosing this field.

.....

.....[2]

(c) State the data type you would choose for each of the following fields.

Garage

Number of Bedrooms

Price in \$[3]

9.1 Database

Answer 6 :

(a) – 7 [1]

(b) – Brochure No
– Uniquely identifies each property [2]

(c) Garage – Boolean
Number of Bedrooms – Number/Integer/Single
Price in \$ – Number/Single/Real/Currency [3]

9.1 Database

Question 7 :

A database, MARKS, was set up to record the test results for a class of students. Part of the database is shown below.

Student Name	Class ID	Maths	English	Science	History	Geography
Paul Smith	0017	70	55	65	62	59
Ravi Gupta	0009	29	34	38	41	44
Chin Hwee	0010	43	47	50	45	52
John Jones	0013	37	67	21	28	35
Diana Abur	0001	92	88	95	89	78
Rosanna King	0016	21	13	11	27	15

- (a) Give the number of fields that are in each record.
.....[1]
- (b) State which field you would choose for the primary key.
.....
Give a reason for choosing this field.
.....
.....[2]
- (c) – Using the database table , write the structured query language (SQL) query to return Math's **below than** 30 and the **students name** and **output** what would be produced by the structured query language (SQL) statement: [4]

9.1 Database

Answer 7 :

(a) – 7 [1]

(b) – Class ID [2]
– Uniquely identifies each student

9.1 Database

Question 8 :

A database was set up to show the properties of certain chemical elements. Part of the database is shown below.

Name of element	Element symbol	Atomic number	Atomic weight	Melting point (C)	Boiling point (C)	State at room temp
oxygen	O	8	16	−218	−183	gas
iron	Fe	26	56	1538	2861	solid
mercury	Hg	80	201	−38	356	liquid
bromine	Br	35	80	−7	59	liquid
osmium	Os	76	190	3033	5012	solid
caesium	Cs	55	133	28	671	solid
gallium	Ga	31	70	30	2204	solid
argon	Ar	18	40	−189	−186	gas
silver	Ag	47	108	961	2162	solid

(a) How many fields are in each record?
..... [1]

Which field would be best suited as primary key?

(b) [1]

(c) - Using the database table , write the structured query language (SQL) query to return **Atomic number greater than 50** and give the **state at room temp** and **name of element** and give the **output** that would be produced by the structured query language (SQL) statement: [4]

9.1 Database

Answer 8 :

(a) 7 [1]

(b) Element symbol [1]

9.1 Database

Question 9 :

A database, THEATRETOURS, was set up to show the tour dates, towns, number of seats and prices in local currency for a Shakespeare play.

Town	Tour Date	Number of Seats	Price Local Currency
Wigan	18/08/2016	120	15.00
Dumfries	20/08/2016	160	12.50
Turin	25/08/2016	200	17.00
Macon	27/08/2016	75	18.00
Bordeaux	29/08/2016	170	20.00
Algiers	01/09/2016	125	1350.00
Windhoek	05/09/2016	65	90.00
Windhoek	06/09/2016	65	90.00
Port Elizabeth	10/09/2016	200	110.00

(a) Explain why none of the fields in the database can be used as a primary key.

.....

.....

.....[2]

9.1 Database

Question 9 :

(b) State a field that could be added as a primary key.

.....

Give a reason for choosing this field.

.....

.....[2]

(c) - Complete the structured query language (SQL) query to return the **town**, **tour date** and **number of seat** in the database table and give the **output** that would be produced by the structured query language (SQL) statement:

- **SELECT**,,
- **FROM**
- **WHERE** = 125;

[4]

9.1 Database

Answer 9 :

- (a) – Town has duplicate entries/all fields can have duplicate entries [2]
– fields other than Town not suitable identifiers

- (b) – Performance number ... [2]
– ... uniquely identifies each performance

(c) – output=

Algiers	01/09/2016	125
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[4]

9.1 Database

Question 10 :

The database table, PCSTOCK, is a part of the database in an electronics shop, showing some of the desktop (DT), tablet (TB) and laptop (LT) computers they have in stock.

PCID	ScreenSize	RAM	Type	HDD(GB)	Price
DT303240	30	32	DT	4000	\$5000.00
DT303220	30	32	DT	2000	\$4500.00
DT301620	30	16	DT	2000	\$4000.00
DT231610	23	16	DT	1000	\$3000.00
LT191620	19	16	LT	2000	\$3000.00
LT171610	17	16	LT	1000	\$2500.00
DT230820	23	8	DT	2000	\$2000.00
DT190810	19	8	DT	1000	\$1500.00
LT190810	19	8	LT	1000	\$1500.00
LT170805	17	8	LT	500	\$1200.00
DT230420	23	4	DT	2000	\$1000.00
DT190410	19	4	DT	1000	\$750.00
LT190410	19	4	LT	1000	\$950.00
TB100206	10	2	TB	64	\$200.00

- (a) Complete the table to show the most appropriate data type for each field based on the data shown in the table at the start of question 6.

Field	Data type
PCID	
ScreenSize	
Type	
Price	

[2]

9.1 Database

Question 10 :

(b)- Give the output that would be produced by the structured query language (SQL) statement:

```
SELECT PCID, ScreenSize, RAM, Type, HDD(GB), Price  
FROM , PCSTOCK  
WHERE HDD(GB) = 500;
```

9.1 Database

Answer 10 :

(a)

Field	Data type
PCID	Text
ScreenSize	Number
Type	Text
Price	Currency

2 marks for 4 correct data types

1 mark for 2 or 3 correct data types

2

(b)-

LT170805	17	8	LT	500	\$1200.00
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